

The sharp mixed-norm threshold for the Pure Greedy Algorithm

Solution packet for arXiv:2304.06423

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Candidate full answer. The rate $\gamma_m(\alpha, H) = O(m^{-\alpha/2})$ holds precisely for $0 < \alpha \leq \alpha_*$, where

$$\alpha_* = 0.365551412606085 \dots$$

It therefore extends strictly beyond $1/3$, but fails for every $\alpha > \alpha_*$.

1 The question

Let H be a real Hilbert space and let $\mathcal{D} \subset H$ be a symmetric dictionary of unit vectors with dense linear span. For $f \in H$, the Pure Greedy Algorithm (PGA) starts with $r_0 = f$ and iterates

$$d_m = \sup_{g \in \mathcal{D}} |\langle r_m, g \rangle|, \quad r_{m+1} = r_m - d_m g_{m+1},$$

where the sign and $g_{m+1} \in \mathcal{D}$ are chosen so that $\langle r_m, g_{m+1} \rangle = d_m$. Write $\|f\|_{A_1(\mathcal{D})}$ for the gauge of the closed convex hull of $\mathcal{D}$, and define

$$\gamma_m(\alpha, H) = \sup_{\mathcal{D}, f, \text{PGA}} \frac{\|r_m\|}{\|f\|^{1-\alpha} \|f\|_{A_1(\mathcal{D})}^\alpha}, \quad 0 < \alpha \leq 1.$$

The source, V. N. Temlyakov, *On the rate of convergence of greedy algorithms*, arXiv:2304.06423v1 [1], proves

$$\frac{1}{2} m^{-\alpha/2} \leq \gamma_m(\alpha, H) \leq m^{-\alpha/2}, \quad 0 < \alpha \leq \frac{1}{3}, \quad (1)$$

and asks on PDF page 5 whether the upper rate continues to hold for $\alpha > 1/3$. It notes, using the lower exponent known at the time, that it cannot continue beyond 0.3796.

We do not know if the upper bound in (1.4) holds for $\alpha > 1/3$. However, the inequality in (1.3) and the lower bound for the $\gamma_m(H)$ show that

$$\gamma_m(\alpha, H) \geq \gamma_m(H) \geq c m^{-0.1898}.$$

Therefore, the upper bound in (1.4) cannot be extended beyond $\alpha_0 := 0.3796$.

Section 3 deals with the case of a Banach space X . Results for the Banach space case are similar to those for Hilbert spaces but are not as sharp as their counterparts.

2 Hilbert space The Weak Greedy Algorithm

Figure 1: The exact question and the source's original obstruction on page 5 of arXiv:2304.06423v1.

We determine the exact transition point. Let $\Gamma \in [1, 3/2]$ be the unique root of

$$(1+x)^{1/(2+x)} \left(1 + \frac{1}{1+x}\right) - 1 - \frac{1}{x} = 0. \quad (2)$$

Numerically,

$$\Gamma = 1.1523436882652335\dots, \quad \alpha_* := \frac{\Gamma}{2+\Gamma} = 0.3655514126060854\dots$$

2 Proof intuition

The familiar PGA proof tracks two quantities: residual energy $a_m = \|r_m\|^2$ and accumulated atomic mass $b_m = \|f\|_{A_1(\mathcal{D})} + \sum_{j < m} d_j$. One elementary recurrence gives $a_m b_m^{-2} \lesssim m^{-1}$. Sil'nichenko's sharp standard-rate proof contains a second invariant, customarily normalized with $\|f\|_{A_1(\mathcal{D})} = 1$. Its scalar differential inequality is actually invariant under *independent* rescaling of the horizontal mass variable by $\|f\|_{A_1(\mathcal{D})}$ and the vertical energy variable by $\|f\|^2$. Restoring both scales gives

$$a_m b_m^\Gamma \lesssim \|f\|^2 \|f\|_{A_1(\mathcal{D})}^\Gamma.$$

Taking the right powers of these two invariants cancels b_m and gives the desired mixed estimate at α_* . Interpolation with the trivial bound $\|r_m\| \leq \|f\|$ covers all smaller α . The sharp matching-pursuit lower construction of Klusowski–Siegel then forbids the rate for every larger α .

3 The two-scale envelope lemma

We record the part of Sil'nichenko's argument that carries the new scaling.

Lemma 1 (two-scale Sil'nichenko invariant). *Let r_m be a PGA residual sequence, put*

$$h = \|f\|, \quad A = \|f\|_{A_1(\mathcal{D})}, \quad a_m = \|r_m\|^2, \quad b_0 = A, \quad b_{m+1} = b_m + d_m.$$

With Γ defined by (2), there is an absolute constant C_Γ such that

$$a_m b_m^\Gamma \leq C_\Gamma h^2 A^\Gamma \quad (m \geq 0). \quad (3)$$

Proof. We expose the scaling in the scalar proof of [2]. The PGA identities and the atomic norm give

$$a_{m+1} = a_m - d_m^2, \quad \|r_m\|_{A_1(\mathcal{D})} \leq b_m. \quad (4)$$

For $\ell \geq k$, the definition of d_ℓ and $\|r_k\|_{A_1(\mathcal{D})} \leq b_k$ imply

$$d_\ell b_k \geq |\langle r_\ell, r_k \rangle|.$$

Since $r_\ell = r_k - \sum_{j=k}^{\ell-1} d_j g_{j+1}$ and $|\langle r_k, g_{j+1} \rangle| \leq d_k$, it follows that

$$d_\ell \geq \frac{a_k - d_k(b_\ell - b_k)}{b_k} \quad (5)$$

whenever the numerator is positive; if it is nonpositive the inequality is automatic.

Interpolate a_m and b_m linearly in the iteration variable. On an interval corresponding to step m , (4) gives $a' = -d_m^2$ and $b' = d_m$. Regarding energy as a function F of accumulated mass, so $F(b(t)) = a(t)$, gives $F' = -d_m$. Inequality (5), first at knots and then by linear interpolation, becomes the scalar envelope inequality

$$F'(x_2) \leq \frac{-F(x_1) - F'(x_1)(x_2 - x_1)}{x_1}, \quad A \leq x_1 \leq x_2. \quad (6)$$

The barrier lemma proved in [2] states that a nonnegative decreasing piecewise-linear function Φ on $[1, \infty)$ satisfying (6) and $\Phi(1) \leq 1$ obeys

$$\Phi(x)x^\Gamma \leq C_\Gamma, \quad x \geq 1, \quad (7)$$

where the largest admissible exponent Γ is characterized by (2). For completeness, its proof sets $x = e^t$ and $G(t) = e^{\Gamma t}\Phi(e^t)$; applying (6) at the optimizing delay $t_0 = \log(1 + \Gamma)/(2 + \Gamma)$ gives a delayed linear inequality for G . The equality case for the barrier is exactly (2), and integration over consecutive intervals of length t_0 bounds G uniformly. The same conclusion up to the right endpoint holds when the domain is a finite interval, by applying the argument only to intervals that precede that endpoint. Thus no assumption that $b_m \rightarrow \infty$ is needed.

The key point is that no relation between the two initial scales was used in this scalar lemma. Define

$$\Phi(z) = \frac{F(Az)}{h^2}, \quad z \geq 1.$$

Then $\Phi(1) = 1$, and direct substitution shows that (6) is unchanged: both sides acquire the same factor A/h^2 . Applying (7) at $z = b_m/A$ gives

$$\frac{a_m}{h^2} \left(\frac{b_m}{A} \right)^\Gamma \leq C_\Gamma,$$

which is (3). $\square$

4 Sharp upper range

The other invariant is elementary.

Lemma 2. *For every $m \geq 0$,*

$$a_m b_m^{-2} \leq (m+1)^{-1}. \quad (8)$$

Proof. The atomic duality estimate gives $d_m \geq a_m/\|r_m\|_{A_1(\mathcal{D})} \geq a_m/b_m$. Hence, with $x_m = a_m b_m^{-2}$ and using $b_{m+1} \geq b_m$,

$$x_{m+1} \leq x_m(1 - x_m).$$

Also $x_0 = h^2/A^2 \leq 1$. The standard reciprocal induction yields $x_m \leq (m+1)^{-1}$. $\square$

Theorem 3 (sharp upper endpoint). *There is an absolute constant C such that, for every Hilbert space, dictionary, target, realization of the PGA, and $m \geq 1$,*

$$\|r_m\| \leq C \|f\|^{1-\alpha_*} \|f\|_{A_1(\mathcal{D})}^{\alpha_*} m^{-\alpha_*/2}. \quad (9)$$

Consequently, for every $0 < \alpha \leq \alpha_$,*

$$\gamma_m(\alpha, H) \leq C_\alpha m^{-\alpha/2}. \quad (10)$$

Proof. Raise (3) to the power $2/(2 + \Gamma)$ and (8) to the power $\Gamma/(2 + \Gamma)$, and multiply. The powers of b_m cancel and give

$$a_m \leq C_\Gamma^{2/(2+\Gamma)} h^{4/(2+\Gamma)} A^{2\Gamma/(2+\Gamma)} (m+1)^{-\Gamma/(2+\Gamma)}.$$

Taking square roots and using $\alpha_* = \Gamma/(2 + \Gamma)$ proves (9).

For $0 < \alpha < \alpha_*$, put $\lambda = \alpha/\alpha_*$. Energy decreases, so $\|r_m\| \leq h$. The elementary inequality $\min\{u, v\} \leq u^{1-\lambda} v^\lambda$ applied to this trivial bound and (9) proves (10). $\square$

The orthonormal example already used in the source shows that this rate is optimal throughout the upper range. Namely, take $f = \sum_{j=1}^{2m} e_j$ and the signed orthonormal dictionary. After m steps, $\|r_m\| = m^{1/2}$, while $\|f\| = (2m)^{1/2}$ and $\|f\|_{A_1(\mathcal{D})} = 2m$. Thus

$$\gamma_m(\alpha, H) \geq 2^{-(1+\alpha)/2} m^{-\alpha/2} \geq \frac{1}{2} m^{-\alpha/2}. \quad (11)$$

5 Failure above the threshold

Klusowski and Siegel [3] proved the sharp lower exponent for the standard PGA problem. In the present notation, if

$$p_* := \frac{\Gamma}{2(2+\Gamma)} = \frac{\alpha_*}{2},$$

then for every $p > p_*$ there are a fixed Hilbert space, a fixed dictionary, and a fixed nonzero $f \in A_1(\mathcal{D})$ whose PGA residuals satisfy

$$\|r_m\| \geq c_p \|f\|_{A_1(\mathcal{D})} m^{-p} \quad (12)$$

for all sufficiently large m .

Theorem 4 (sharp failure). *If $\alpha > \alpha_*$, there is no constant C_α for which $\gamma_m(\alpha, H) \leq C_\alpha m^{-\alpha/2}$ holds uniformly in m .*

Proof. Choose p with $p_* < p < \alpha/2$ and take the fixed example in (12). Since $h = \|f\|$ and $A = \|f\|_{A_1(\mathcal{D})}$ are fixed positive numbers,

$$\gamma_m(\alpha, H) \geq \frac{c_p A}{h^{1-\alpha} A^\alpha} m^{-p} = c'_{p,\alpha} m^{-p}.$$

Multiplying by $m^{\alpha/2}$ tends to infinity, which contradicts any uniform target bound. $\square$

Corollary 5 (complete answer to the source question). *The upper rate in (1) extends beyond $1/3$ exactly through*

$$\alpha_* = \frac{\Gamma}{2+\Gamma} = 0.365551412606085 \dots$$

For $0 < \alpha \leq \alpha_*$ its order is exactly $m^{-\alpha/2}$; for every $\alpha > \alpha_*$ that order is impossible uniformly over dictionaries and targets.

6 Scope, novelty, and review

The result concerns the order of the uniform rate; constants may depend on α . It does not assert that the unit constant in the source’s display remains optimal beyond $1/3$. The threshold comes from combining a new two-norm reading of Sil’nichenko’s upper proof with the later sharp lower construction. Neither supporting paper states the mixed characteristic $\gamma_m(\alpha, H)$ at this threshold.

The run indexes, parsed corpus, exact formula, source question, and bounded primary-source web searches through 11 August 2026 found no explicit prior statement of this characterization. Temlyakov’s 2025 survey records the sharp standard exponent but not the two-scale invariant or mixed threshold. Novelty confidence is moderate because the rescaling is short and may be implicit folklore.

The proof is analytic. The only numerical value is an evaluation of the uniquely characterized root and is not used as evidence. Expert review should focus on the passage from the discrete cross-iteration inequality to the scalar envelope inequality, the independent (A, h^2) rescaling in Lemma 1, and the quantifiers in the Klusowski–Siegel lower theorem.

References

- [1] V. N. Temlyakov, *On the rate of convergence of greedy algorithms*, Mathematics **11** (2023), 2559; arXiv:2304.06423v1.
- [2] A. V. Sil’nichenko, *Rate of convergence of greedy algorithms*, Math. Notes **76** (2004), 582–586.
- [3] J. M. Klusowski and J. W. Siegel, *Sharp convergence rates for matching pursuit*, IEEE Trans. Inform. Theory **71** (2025), 5556–5572; arXiv:2307.07679.